

Importing the Dependencies

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn import metrics
```

Data Collection & Analysis

```
# loading the data from csv file to a Pandas DataFrame
insurance_dataset = pd.read_csv('/content/insurance.csv')
```

```
# first 5 rows of the dataframe
insurance_dataset.head()
```

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520

Next steps:

[Generate code with insurance_dataset](#)
[View recommended plots](#)
[New interactive sheet](#)

```
# number of rows and columns
insurance_dataset.shape
```

```
(1338, 7)
```

```
# getting some informations about the dataset
insurance_dataset.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1338 entries, 0 to 1337
Data columns (total 7 columns):
#   Column      Non-Null Count  Dtype
---  -
0   age         1338 non-null   int64
1   sex         1338 non-null   object
2   bmi         1338 non-null   float64
3   children    1338 non-null   int64
4   smoker      1338 non-null   object
5   region      1338 non-null   object
6   charges     1338 non-null   float64
dtypes: float64(2), int64(2), object(3)
memory usage: 73.3+ KB
```

Categorical Features:

- Sex
- Smoker
- Region

```
# checking for missing values
insurance_dataset.isnull().sum()
```



```

0
age    0
sex    0
bmi    0
children 0
smoker 0
region 0
charges 0

```

dtype: int64

Data Analysis

```

# statistical Measures of the dataset
insurance_dataset.describe()

```



	age	bmi	children	charges
count	1338.000000	1338.000000	1338.000000	1338.000000
mean	39.207025	30.663397	1.094918	13270.422265
std	14.049960	6.098187	1.205493	12110.011237
min	18.000000	15.960000	0.000000	1121.873900
25%	27.000000	26.296250	0.000000	4740.287150
50%	39.000000	30.400000	1.000000	9382.033000
75%	51.000000	34.693750	2.000000	16639.912515
max	64.000000	53.130000	5.000000	63770.428010



```

# distribution of age value
sns.set()
plt.figure(figsize=(6,6))
sns.distplot(insurance_dataset['age'])
plt.title('Age Distribution')
plt.show()

```



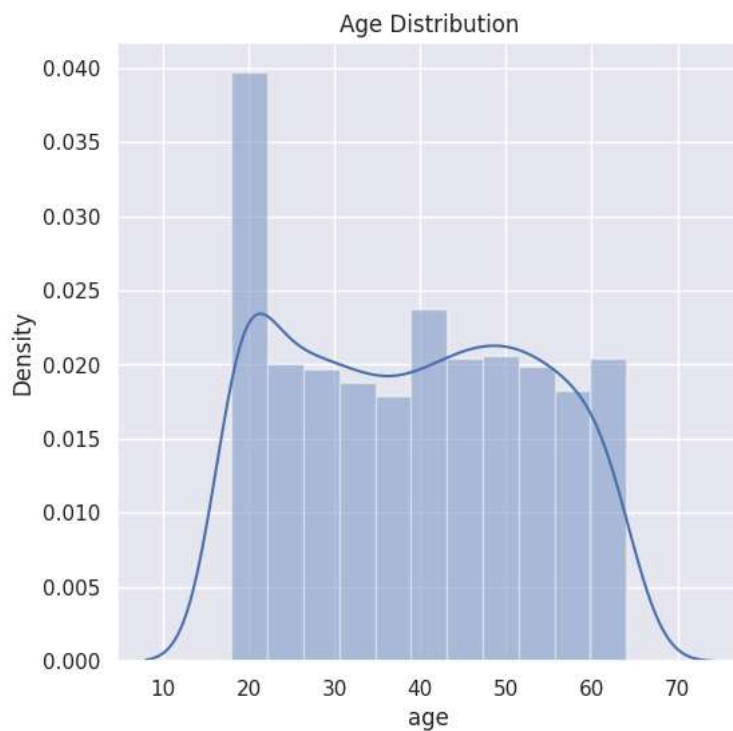
<ipython-input-8-28228e9c3528>:4: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

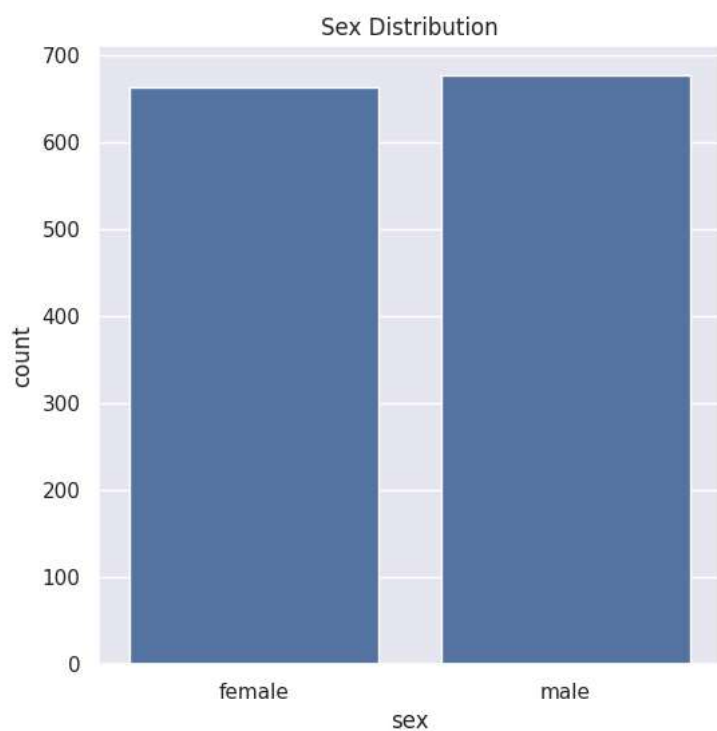
Please adapt your code to use either `displot` (a figure-level function with similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see <https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751>

```
sns.distplot(insurance_dataset['age'])
```



```
# Gender column  
plt.figure(figsize=(6,6))  
sns.countplot(x='sex', data=insurance_dataset)  
plt.title('Sex Distribution')  
plt.show()
```



```
insurance_dataset['sex'].value_counts()
```



count	
sex	
male	676
female	662

dtype: int64

```
# bmi distribution
plt.figure(figsize=(6,6))
sns.distplot(insurance_dataset['bmi'])
plt.title('BMI Distribution')
plt.show()
```



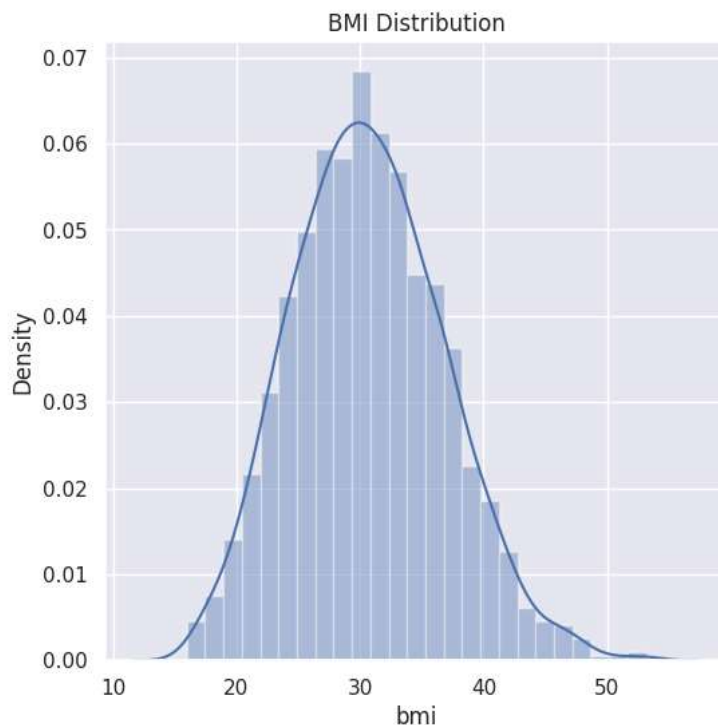
<ipython-input-11-81b69896b0d5>:3: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with similar flexibility) or `histplot` (an axes-level function for histograms).

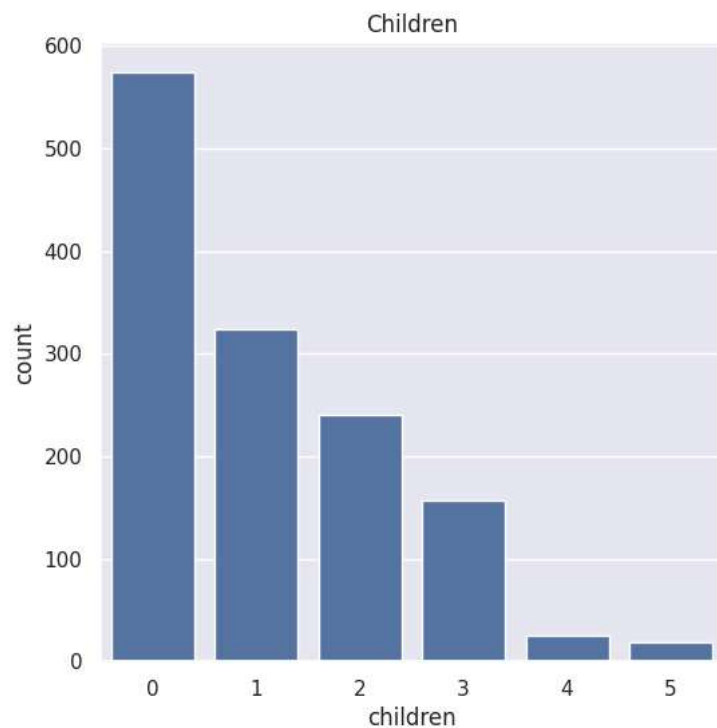
For a guide to updating your code to use the new functions, please see <https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751>

```
sns.distplot(insurance_dataset['bmi'])
```



Normal BMI Range --> 18.5 to 24.9

```
# children column
plt.figure(figsize=(6,6))
sns.countplot(x='children', data=insurance_dataset)
plt.title('Children')
plt.show()
```



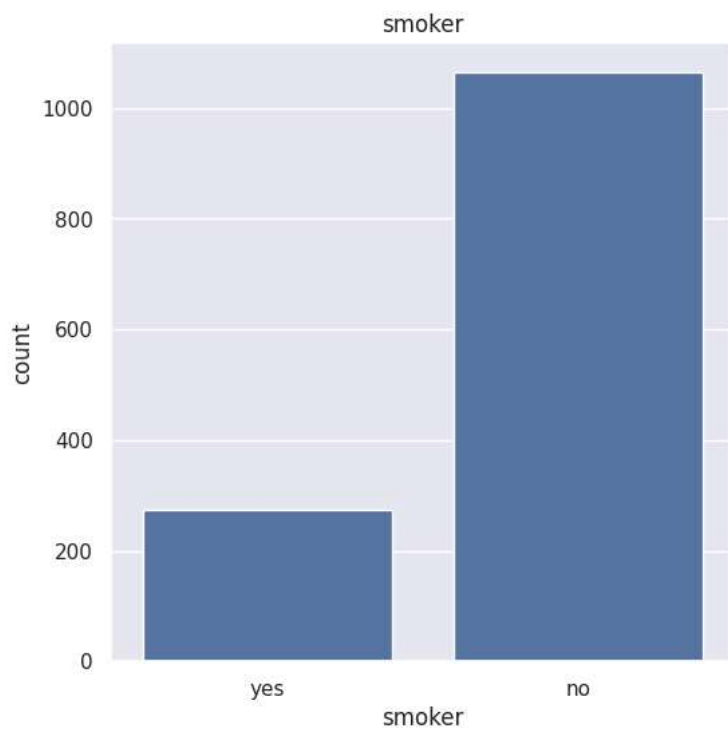
```
insurance_dataset['children'].value_counts()
```



	count
children	
0	574
1	324
2	240
3	157
4	25
5	18

dtype: int64

```
# smoker column
plt.figure(figsize=(6,6))
sns.countplot(x='smoker', data=insurance_dataset)
plt.title('smoker')
plt.show()
```



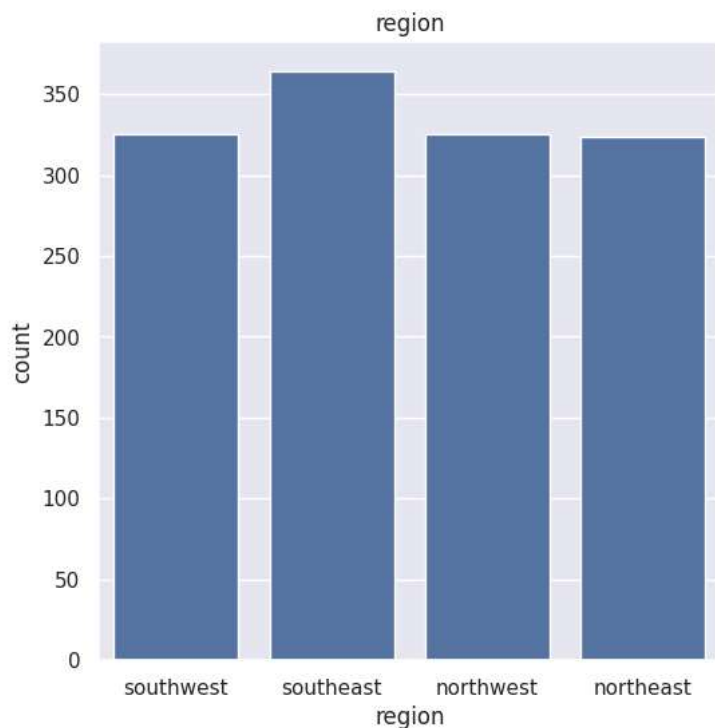
```
insurance_dataset['smoker'].value_counts()
```



	count
no	1064
yes	274

dtype: int64

```
# region column
plt.figure(figsize=(6,6))
sns.countplot(x='region', data=insurance_dataset)
plt.title('region')
plt.show()
```



```
insurance_dataset['region'].value_counts()
```

	count
region	
southeast	364
southwest	325
northwest	325
northeast	324

dtype: int64

```
# distribution of charges value
plt.figure(figsize=(6,6))
sns.distplot(insurance_dataset['charges'])
plt.title('Charges Distribution')
plt.show()
```

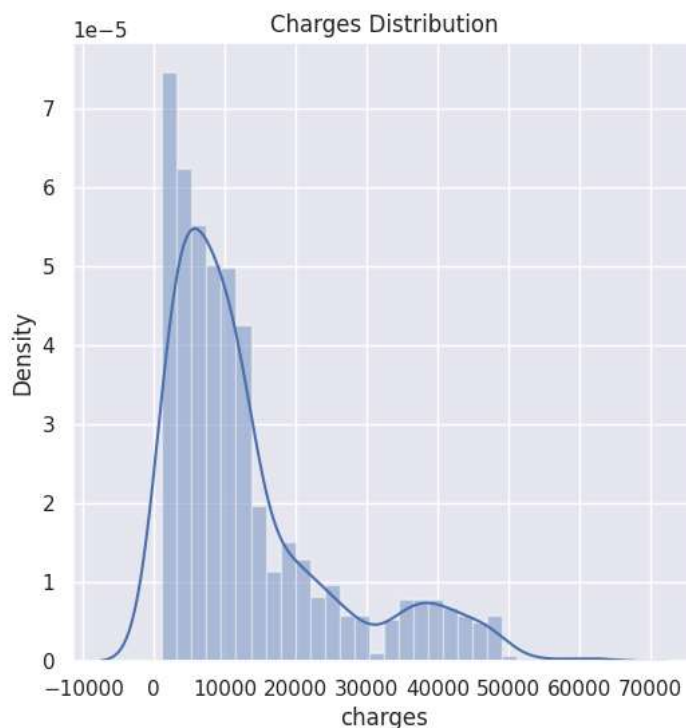
<ipython-input-19-a2fe9b394a51>:3: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see <https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751>

```
sns.distplot(insurance_dataset['charges'])
```



Data Pre-Processing

Encoding the categorical features

```
# encoding sex column
insurance_dataset.replace({'sex':{'male':0,'female':1}}, inplace=True)
```

```
3 # encoding 'smoker' column
insurance_dataset.replace({'smoker':{'yes':0,'no':1}}, inplace=True)
```

```
# encoding 'region' column
insurance_dataset.replace({'region':{'southeast':0,'southwest':1,'northeast':2,'northwest':3}}, inplace=True)
```

<ipython-input-20-7d5826986d65>:2: FutureWarning: Downcasting behavior in `replace` is deprecated and will be removed in a future version of pandas. To suppress this warning, please call `pandas.set\_option('future.no\_downcasting', 1)` before using `replace`.

```
insurance_dataset.replace({'sex':{'male':0,'female':1}}, inplace=True)
```

<ipython-input-20-7d5826986d65>:5: FutureWarning: Downcasting behavior in `replace` is deprecated and will be removed in a future version of pandas. To suppress this warning, please call `pandas.set\_option('future.no\_downcasting', 1)` before using `replace`.

```
insurance_dataset.replace({'smoker':{'yes':0,'no':1}}, inplace=True)
```

<ipython-input-20-7d5826986d65>:8: FutureWarning: Downcasting behavior in `replace` is deprecated and will be removed in a future version of pandas. To suppress this warning, please call `pandas.set\_option('future.no\_downcasting', 1)` before using `replace`.

```
insurance_dataset.replace({'region':{'southeast':0,'southwest':1,'northeast':2,'northwest':3}}, inplace=True)
```

Splitting the Features and Target

```
X = insurance_dataset.drop(columns='charges', axis=1)
Y = insurance_dataset['charges']
```

```
print(X)
```

```

↗
   age  sex    bmi  children  smoker  region
0    19   1  27.900         0       0       1
1    18   0  33.770         1       1       0
2    28   0  33.000         3       1       0
3    33   0  22.705         0       1       3
4    32   0  28.880         0       1       3
...  ...  ...    ...      ...    ...    ...
1333  50   0  30.970         3       1       3
1334  18   1  31.920         0       1       2
1335  18   1  36.850         0       1       0
1336  21   1  25.800         0       1       1
1337  61   1  29.070         0       0       3

```

```
[1338 rows x 6 columns]
```

```
print(Y)
```

```

↗
0    16884.92400
1    1725.55230
2    4449.46200
3    21984.47061
4    3866.85520
...
1333  10600.54830
1334  2205.98080
1335  1629.83350

```